

SQL + JOIN CARDINALITY

Valid SQL. Wrong dataset.

What happened when I joined
MusicBrainz artists on name?

2.93M

rows before JOIN

5.38M

rows after JOIN

+83.52% without a syntax error

Why 249 entities became 62,001 rows.

$$249 \times 249 = 62,001$$

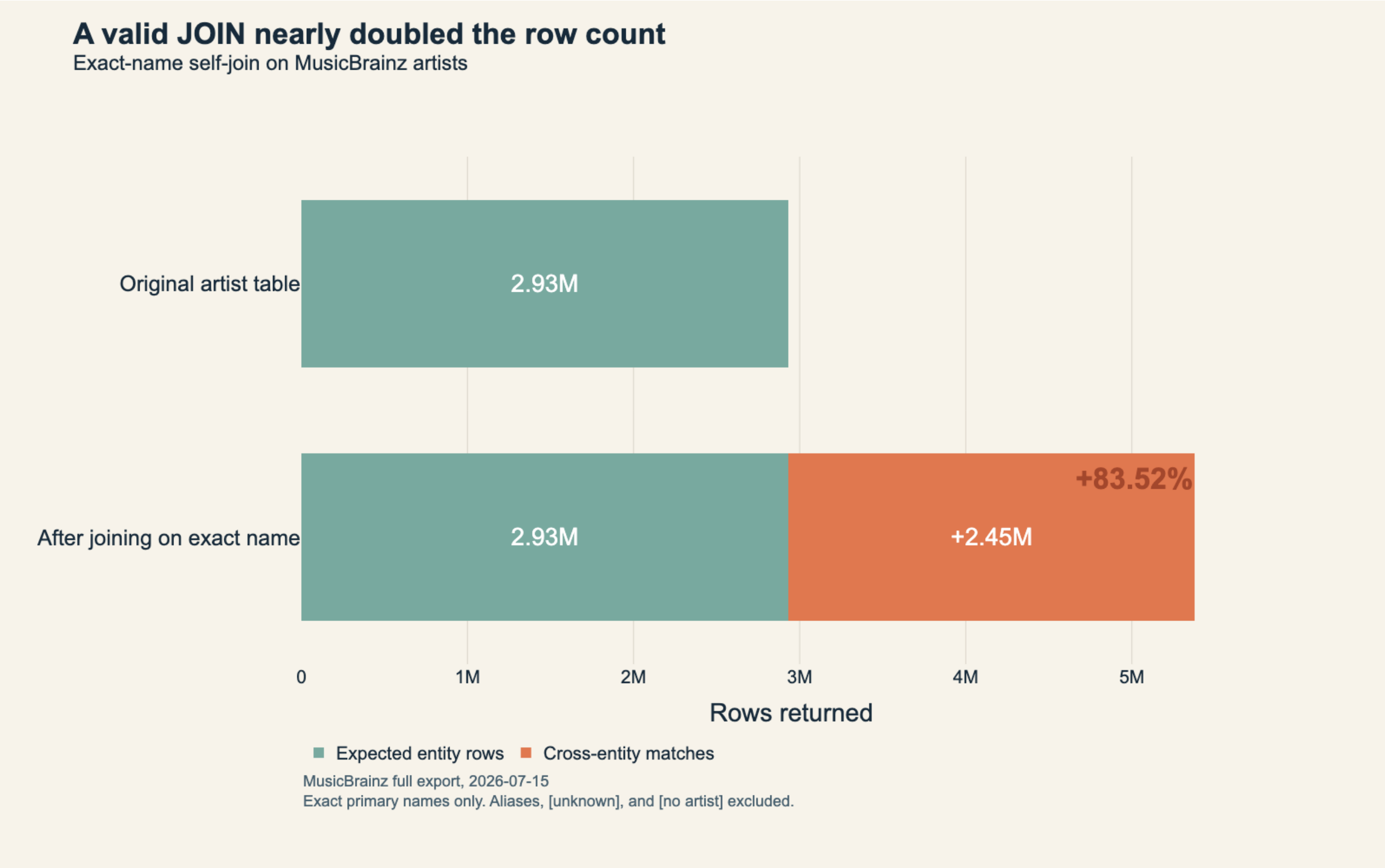
Every Indigo row matched every other Indigo row.

249
same-ID matches

61,752
cross-entity matches

The query ran correctly. The join key created the wrong grain.

The extra rows came from different artist IDs.



2,448,276 cross-entity matches.

Three questions before every JOIN.

1

GRAIN

What does one row represent in each input?

2

KEY

Does the column identify an entity or only describe it?

3

CARDINALITY

How many rows should one input row match?

```
artist.name = descriptive attribute  
artist.id / artist.gid = catalog identifiers
```


What this analysis does not prove.

- This was a controlled self-join, not a production incident.
- Exact primary names only. Aliases were excluded.
- [unknown] and [no artist] were excluded.
- Different catalog IDs do not prove different real-world people.

NEXT EPISODE

Why is a track not always the same as a recording?

MusicBrainz SQL Stories #3